

### Solve the given problems:

1. Compute  $y(1.2)$  from the following IVPs

(i)  $y' = x + y, y(1) = 0,$

(ii)  $y' = -y^2, y(1) = 1,$

using (a) Euler method, (b) Modified Euler method, (c) Euler-Cauchy method, and (d) R-K method of fourth order (classical method) with  $h = 0.1$  in each case. Find the errors in each case.

2. Discretize the following IVP

$$y' = 1 + a(1 + x - y), y(0) = 2,$$

using Euler method with step length  $h$ .

Solve the IVP to estimate  $y(1)$  using  $h = 0.5$  and  $h = 0.25$  with  $a = 0.25$ .

Find the errors in each case.

3. Use the Taylor series method of four order to solve the following IVP

$$y' = x^2 + y^2, y(0) = 1,$$

For the interval  $(0, 0.4)$  with step length  $h = 0.2$ .

4. Given the equation

$$y' = x + \sin y, y(0) = 1$$

show that it is sufficient to use Euler method with the step  $h = 0.2$  to compute  $y(0.2)$  with an error less than 0.05.

5. Apply the Euler Cauchy method with step length  $h$  to the differential equation  $y' = 1 + a(1 + x - y)$ ,  $a > 0$ , with the general solution  $y(x) = 1 + x + ce^{-ax}$ , where  $c$  is an arbitrary constant.
- Determine an explicit expression for  $y_n$ .
  - For which values of  $h$  is the sequence  $\{y_n\}_{n=0}^{\infty}$  bounded.
  - Estimate  $y(0.4)$  numerically using the condition  $y(0) = 1$ , with  $a = 0.5$  and  $h = 0.2$ .
6. Use the classical Runge-Kutta method of fourth order to find the numerical solution at  $x = 0.8$  for  $y' = \sqrt{x + y}$ ,  $y(0.4) = 0.41$ . Assume the step length  $h = 0.2$ .
7. Determine  $y(x)$  at  $x = 0.2, 0.4$ , using the classical fourth order Runge-Kutta method, given that  $y' = \frac{x}{y}$ ,  $y(0) = 1$ . Compare the results with the exact solution. Find the error at  $t = 0.4$ .
8. The IVP is given by  $y' = xy$ ,  $y(0) = 1$ . Estimate the error at  $x = 1$ , when Euler method is used, with step size  $h = 0.01$ .

Use the error formula

$$|y(x_j) - y_j| \leq \frac{hM}{2L} \left( \exp((x_j - a)L) - 1 \right), \text{ when Euler's method is applied to the}$$

problem  $y' = f(x, y)$ ;  $y(a) = A$ , in  $a \leq x \leq b$  and  $h = \frac{b-a}{N}$ ,  $x_j = a + jh$  and

$$\left| \frac{\partial f}{\partial y} \right| \leq L, \quad |y''(x)| \leq M.$$

9. Apply the Modified Euler method with step length  $h$  to the IVP

$$y' = f(x, y), \quad y(x_0) = y_0, \quad x \in [x_0, b].$$

(i) Find the local truncation and global error.

(ii) Is this numerical method consistent?

(iii) What is the order of the method? Justify.

(iv) Solve the IVP  $y' = \frac{y^2 - x^2}{y^2 + x^2}$ ,  $y(0) = 1$  to find  $y(0.1)$  using this method with step length  $h = 0.1$ .

10. Apply the Trapezium method with step length  $h$  to the IVP

$$y' = f(x, y), \quad y(x_0) = y_0, \quad x \in [x_0, b].$$

(i) Find the local truncation and global error.

(ii) Is this numerical method consistent?

(iii) What is the order of the method? Justify.

(vi) Solve the IVP  $y' = \frac{y - x^2}{y + x^2}$ ,  $y(0) = 1$  to find  $y(0.1)$  using this method with step length  $h = 0.1$ .